

YouTube Channel Analytics Dashboard

1. Introduction

This document summarizes the development of a **YouTube Channel Analytics Dashboard** built with Streamlit inside Snowflake Notebooks.

The main objective was to create an interactive and user-friendly web application that allows YouTube creators to easily track and analyze their channel performance over a 5-year period directly within the Snowflake environment.

The dashboard provides clear visualizations of key metrics with flexible filtering options and dynamic charts.

2. Environment Setup

The project was developed entirely inside **Snowflake Notebooks** using the following:

- **Platform:** Snowflake Notebooks (with native Streamlit support)
- **Language:** Python 3
- **Libraries:** Streamlit, Pandas, NumPy
- **Time Period:** August 2019 – September 2024 (62 months)

No local installation was required — everything was built and run directly in Snowflake.

3. Data Generation

Realistic synthetic monthly data was generated to simulate actual YouTube channel growth.

The data includes:

- Steady linear growth in subscribers and views
- Random variations for realism
- Seasonal patterns (higher activity in summer)
- Occasional viral spikes (every 6 months on average)

Key columns created: Date, Net Subscribers, Views, Watch Hours, Likes, Shares, and Comments.

4. Dashboard Features

- Date Range Selector
- Time Frame Options: Daily, Weekly, Monthly, Quarterly
- Chart Type Selection: Bar Chart or Area Chart
- Key Performance Metrics with growth indicators (Net Subscribers, Views, Watch Hours, Likes)

- Dynamic Charts that update based on filters
- Yearly Summary Table with built-in visualizations (Progress bars, Bar charts, Line charts)
- Expandable Raw Data view

5. Dashboard

- **Filter Controls**
Users can select start/end date, time frame, and chart type.

Start date

2019/08/01

End date

2024/09/01

Select time frame

Daily

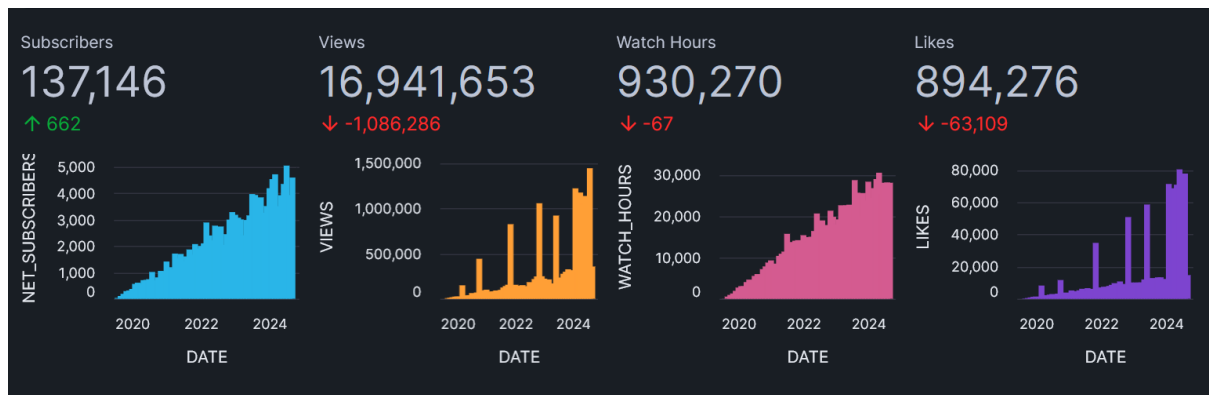
Select a chart type

Bar

- **Yearly Summary Table**
Year-wise aggregated data with embedded visualizations.

YEAR	NET_SUBSCRIBERS	VIEWS	WATCH_HOURS	LIKES	SHARES	COMMENTS
2,019	314					
2,020	1037					
2,021	1908					
2,022	3287					
2,023	3518					
2,024	4599					

- **Main Dashboard View**
Key metrics with growth indicators and dynamic charts.



6. Results & Insights

The dashboard effectively showcases long-term channel growth with clear visual trends.

Key observations:

- Consistent growth in subscribers and watch hours over 5 years
- Significant spikes during simulated viral months
- Visible seasonal effects on views
- Easy comparison across different time periods

This tool helps content creators make data-driven decisions quickly, all within Snowflake.

7. Future Enhancements

- Connect to real YouTube Data API using Snowflake external functions
- Add more metrics (CTR, Average View Duration, Revenue estimation)
- Export reports to PDF or CSV
- Growth forecasting using Snowpark ML
- Share the dashboard as a Snowflake Native App

8. Conclusion

The YouTube Channel Analytics Dashboard successfully demonstrates how to build an interactive Streamlit application directly inside Snowflake Notebooks.

It combines realistic data generation, dynamic visualizations, and a clean user interface — providing practical value for YouTube creators while showcasing modern data app development capabilities in Snowflake.